 MAT 1341 A TEST 5 Soln V3

1. If $C = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, and D is a $3 \times n$ matrix then the second row of the matrix CD is

A. the same as the second row of C.
B. the same as the first row of D.
C. the same as the second row of D.
D. the sum of the first and the second rows of D.
E. the sum of the first and the third rows of D.
F. the sum of the second and third rows of D.

Let $D = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix}$; r_i = ith row of D

Then $CD = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix} = \begin{bmatrix} r_1 + r_2 \\ r_1 + r_3 \\ r_2 \end{bmatrix}$

2. Find all values of x for which $(1,0,1), (-1,2,1)$ and $(1,x,3)$ are linearly independent. (v_1, v_2, v_3)

A. For $x = 2$ only
B. For all x not equal to 2
C. $x = 0$ or $x = 1$
D. For all x
E. For no x
F. For $x = 0$ only

$\{v_1, v_2, v_3\}$ is l.i.o iff rank $[v_1\ v_2\ v_3] = 3$

iff rank $\begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & x \\ 1 & 1 & 3 \end{bmatrix} = 3$

iff rank $\begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & 2 \\ 0 & 2 & x \end{bmatrix} = 3$ iff rank $\begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & x-2 \end{bmatrix} = 3$

iff $x \neq 2$

3. Suppose $A = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$. Which one of the following statements is true ?

A. A^{-1} does not exist.
B. The third row of A^{-1} is $[0 \quad 0 \quad 1]$.
C. The second row of A^{-1} is $[-1 \quad 1 \quad 0]$.
D. The first row of A^{-1} is $[1 \quad 1 \quad -1]$.
E. The second column of A^{-1} is $[0 \quad 1 \quad -1]^t$.
F. All of B, C, D, E are true.

$$[A \mid I_3] = \left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|ccc} 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & -1 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{array}\right] \qquad (A = A^{-1} \text{ here})$$

4. Let

$$A = \begin{bmatrix} 1 & -1 & 1 & 2 \\ 0 & 0 & 1 & 2 \\ -2 & 2 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 1 & 2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 3 & 6 \end{bmatrix}$$

(a) Find a basis for the row space of A.

(b) Find a basis for the column space of A.

(c) Find a basis for $\ker A = \{x \in \mathbf{R}^4 \mid Ax = 0\}$.

(d) Find the dimension of $\{Ax \mid x \in \mathbf{R}^4\}$.

$\sim \begin{bmatrix} \textcircled{1} & -1 & 0 & 0 \\ 0 & 0 & \textcircled{1} & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

(a) By computations above, $\{(1,-1,0,0), (0,0,1,2)\}$ is a basis for row (A).

(b) By computations above $\{(1,0,-2), (1,1,1)\}$ is a basis for col A

(c) $[A|0] \sim \left[\begin{array}{cccc|c} \textcircled{1} & -1 & 0 & 0 & 0 \\ 0 & 0 & \textcircled{1} & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$; $\begin{matrix} x_1 = s \\ x_2 = s \\ x_3 = -2t \\ x_4 = t \end{matrix}$; $s, t \in \mathbb{R}$

$\therefore \ker A = \{ s(1,1,0,0) + t(0,0,-2,1) \mid s, t \in \mathbb{R} \}$

$\therefore \{(1,1,0,0), (0,0,-2,1)\}$ is a basis for $\ker A$

(d) The subspace $\{Ax \mid x \in \mathbb{R}^4\}$ is col A; But

dim col A = rank A = 2 (or see (b)).

5. Let $u_1 = (1,1,0,0), u_2 = (-1,1,0,0), u_3 = (0,0,1,-1)$, and $U = \text{span}\{u_1, u_2, u_3\}$.

a) Show that $\{u_1, u_2, u_3\}$ is an orthogonal set. $\left(U = \{(x,y,z,w) \mid z+w = 0\}\right.$

b) Briefly explain why $\{u_1, u_2, u_3\}$ is a basis of U.

c) Find the best approximation to $(1, -1, 2, -1)$ by vectors in U.

d) Extend $\{u_1, u_2, u_3\}$ to a basis of $\mathbf{R}^4$.

a) A short computation shows that $u_1 \cdot u_2 = u_1 \cdot u_3 = u_2 \cdot u_3 = 0$. Moreover, none of u_1, u_2 or u_3 is zero. Hence $\{u_1, u_2, u_3\}$ is orthogonal.

b) All orthogonal sets are l.i., so $\{u_1, u_2, u_3\}$ is l.i. Moreover, $\{u_1, u_2, u_3\}$ spans U (by def^n) and so $\{u_1, u_2, u_3\}$ is a basis for U.

c) Let $v = (1, -1, 2, -1)$. We seek $\text{proj}_U v$:

$$\text{But } \text{proj}_U v = \frac{v \cdot u_1}{\|u_1\|^2} u_1 + \frac{v \cdot u_2}{\|u_2\|^2} u_2 + \frac{v \cdot u_3}{\|u_3\|^2} u_3$$

$$= 0 + \frac{(-2)}{2} u_2 + \frac{3}{2} u_3$$

$$= \left(1, -1, \tfrac{3}{2}, -\tfrac{3}{2}\right)$$

d) If $w \in \mathbb{R}^4$, then $\{u_1, u_2, u_3, w\}$ is a basis for $\mathbb{R}^4$ iff

$$\text{rank} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ w \end{bmatrix} = 4. \text{ But rank} \begin{bmatrix} 1 & 1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & w & \end{bmatrix} = \text{rank} \begin{bmatrix} \textcircled{1} & 1 & 0 & 0 \\ 0 & \textcircled{2} & 0 & 0 \\ 0 & 0 & \textcircled{1} & -1 \\ & & w & \end{bmatrix}$$

$= 4$ if $w = \{(0,0,0,1)\}$

6(a). State whether the following is true or false, and justify your answer.

If A and B are 4×4 matrices such that the rows of AB are linearly independent, then the columns of A are linearly independent.

TRUE If the rows of AB are linearly independent then AB is invertible. Hence both A & B are invertible. Thus the columns of A are linearly independent

6(b). Let A be a $n \times n$ matrix with real entries. Give three additional statements equivalent to

"The rank of A is n

(i) A is invertible

(ii) $Ax = 0 \Rightarrow x = 0$

(iii) $Ax = b$ is consistent for every $b \in \mathbb{R}^n$.

These are just 3 of a possible 12 answers.

Please consult your notes for others

MAT 134A F14 TEST 5 Solns (VI)

1. If $C = \begin{bmatrix} 0 & 2 & 3 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, and D is a $3 \times n$ matrix then the third row of the matrix CD is

A. the same as the second row of C.
B. the same as the first row of D.
C. the same as the second row of D.
D. the sum of the first and the second rows of C.
E. the sum of twice the first row of D and 3 times the second row of D.
F. the sum of 3 times the first row of D and the second row of D.

Let $D = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix}$ r_i = ith row of D

Then $CD = \begin{bmatrix} 0 & 2 & 3 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix} = \begin{bmatrix} 2r_2 + 3r_3 \\ r_1 + r_3 \\ r_2 \end{bmatrix}$

2. Find all values of x for which $(1,0,2), (-1,0,1)$ and $(1,x,4)$ are linearly independent.

A. For $x = 1$ only
B. For all x not equal to 0
C. $x = 0$ or $x = 1$
D. For all x
E. For no x
F. For $x = 0$ only

These will be l.i. iff rank of $A = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & x \\ 2 & 1 & 4 \end{bmatrix}$ is 3 But

$A \sim \begin{bmatrix} 1 & -1 & 1 \\ 0 & 3 & 2 \\ 0 & 0 & x \end{bmatrix}$ and so rank $A = 3$ iff $x \neq 0$

3. Suppose $A = \begin{bmatrix} 1 & 3 & -1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$. Which one of the following statements is true ?

A. A^{-1} does not exist.
B. The third row of A^{-1} is $[0 \quad 0 \quad 1]$. ×
C. The second row of A^{-1} is $[-2 \quad 1 \quad 0]$.
(D.) The first row of A^{-1} is $[1 \quad 1 \quad -3]$.
E. The second column of A^{-1} is $[0 \quad 2 \quad -1]^t$.
F. All of B, C, D, E are true.

$$\left[\begin{array}{ccc|ccc} 1 & 3 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|ccc} 1 & 3 & -1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{array}\right] \sim \cdots$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & -3 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{array}\right]$$

A^{-1}

4. Let

$$A = \begin{bmatrix} 1 & 2 & -1 & -1 \\ 0 & 0 & 1 & 5 \\ -3 & -6 & 1 & -7 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 & -1 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 2 & -10 \end{bmatrix}$$

(a) Find a basis for the row space of A.

(b) Find a basis for the column space of A.

(c) Find a basis for $\ker A = \{x \in \mathbf{R}^4 \mid Ax = 0\}$.

(d) Find the dimension of $\{Ax \mid x \in \mathbf{R}^4\}$.

$$\sim \begin{bmatrix} 1 & 2 & -1 & -1 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} \textcircled{1} & 2 & 0 & 4 \\ 0 & 0 & \textcircled{1} & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(a) By the computations above, $\{(1,2,0,4), (0,0,1,5)\}$ is a basis for row(A)

(b) By the computation above (using the col space algorithm) $\{(1,0,-3), (-1,1,1)\}$ is a basis for col(A)

(c) $[A|0] \sim \left[\begin{array}{cccc|c} \textcircled{1} & 2 & 0 & 4 & 0 \\ 0 & 0 & \textcircled{1} & 5 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$

$x_1 = -2s - 4t$
$x_2 = s$
$x_3 = -5t$
$x_4 = t$
; $s, t \in \mathbb{R}$

$\therefore \{(-2,1,0,0), (-4,0,-5,1)\}$ is a basis for ker A.

(d) The subspace $\{Ax \mid x \in \mathbb{R}^4\}$ is just col A. But dim col A = rank A = 2. (Or: do (b))

5. Let $u_1 = (1,0,0,1), u_2 = (0,1,0,0), u_3 = (1,0,0,-1)$, and $U = \text{span}\{u_1, u_2, u_3\}$.

a) Show that $\{u_1, u_2, u_3\}$ is an orthogonal set. $\left(U = \{(x,y,z,w) \mid z = 0\}\right)$

b) Briefly explain why $\{u_1, u_2, u_3\}$ is a basis of U.

c) Find the best approximation to $(1, -1, 2, -1)$ by vectors in U.

d) Extend $\{u_1, u_2, u_3\}$ to a basis of $\mathbf{R}^4$.

a) A short computation shows $u_1 \cdot u_2 = u_1 \cdot u_3 = u_2 \cdot u_3 = 0$; Moreover, none of u_1, u_2 or u_3 is zero. Hence $\{u_1, u_2, u_3\}$ is orthogonal.

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b) Every orthogonal set is linearly independent. Since $\{u_1, u_2, u_3\}$ spans U (by def^n of U) and is orthogonal by (a), it is a basis of U.

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c) Let $v = (1, -1, 2, 1)$. We compute $\text{proj}_U v$:

$$\text{proj}_U v = \frac{u_1 \cdot v}{\|u_1\|^2} u_1 + \frac{u_2 \cdot v}{\|u_2\|^2} u_2 + \frac{u_3 \cdot v}{\|u_3\|^2} u_3$$

$$= 0 + \frac{(-1)}{1} \cdot u_2 + \frac{2}{2} u_3 = -u_2 + u_3$$

$$= (1, -1, 0, -1)$$

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c) Let $w \in \mathbb{R}^4$, then $\text{rank} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ w \end{bmatrix} = \text{rank} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 1 \\ w \end{bmatrix}$

$= \text{rank} \begin{bmatrix} \textcircled{1} & 0 & 0 & 1 \\ 0 & \textcircled{1} & 0 & 0 \\ 0 & 0 & 0 & \textcircled{2} \\ w \end{bmatrix} = 4$ if $w = (0,0,1,0)$. Hence $\{u_1, u_2, u_3, w\}$ is a basis for $\mathbb{R}^4$

6(a). State whether the following is true or false, and justify your answer.

If A and B are 4×4 matrices such that the rank of AB is 4, then the rank of A must also be 4.

TRUE : Since rank $AB = 4$, and AB is a 4×4 matrix, AB is invertible. But then we know both A & B are also invertible and so rank $A = 4$.

6(b). Let A be a $n \times n$ matrix with real entries. Give three additional statements equivalent to

"A is invertible"

(i) rank $A = n$

(ii) $A \sim I_n$

(iii) The columns of A are l.i.

are just 3 possible answers. See your notes for 9 other possibilities

7. [Bonus] Suppose W is a subspace of $\mathbf{R}^4$ with $\dim W = 3$. Prove carefully that there is at least one vector $u \in \mathbf{R}^4$, with $||u|| = 1$, and which is orthogonal to every vector in W. How many such vectors are there?

Let $\{w_1, w_2, w_3\}$ be a basis of W. Then

$u \cdot w_1 = u \cdot w_2 = u \cdot w_3 = 0 \iff \underbrace{\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}}_{A} u = 0$ (w_i = i-th row of A)

$\iff u \in \ker A$

($\frac{1}{2}$) Since $\{w_1, w_2, w_3\}$ is l.i., $\operatorname{rank} A = 3$, so $\dim \ker A = 1$.

Let $\{v\}$ be a basis of $\ker A$ and set

$u = \frac{v}{\|v\|}$. Then $u \cdot u_i = 0$, $i = 1, 2, 3$ & $\|u\| = 1$.

($\frac{1}{2}$) There are exactly 2 such vectors: u and $-u$.

MAT B41A FIN TEST V2

1. If $C = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 2 & 3 \end{bmatrix}$, and D is a $3 \times n$ matrix then the third row of the matrix CD is

A. the same as the second row of C.
B. the same as the first row of D.
C. the same as the second row of D.
D. the sum of the first and the second rows of C.
(E.) the sum of twice the second row of D and 3 times the third row of D.
F. the sum of 3 times the first row of D and the second row of D.

Let $D = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix}$, r_i = ith row of D

Then $CD = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 2 & 3 \end{bmatrix}\begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix} = \begin{bmatrix} r_2 \\ r_1 + r_3 \\ 2r_2 + 3r_3 \end{bmatrix}$

2. Find all values of x for which $(-1, 0, 1)$, $(1, 1, 2)$, and $(1, x, 2)$ are linearly independent.
(v_1, v_2, v_3)

A. For $x = 1$ only
(B.) For all x not equal to 1
C. $x = 0$ or $x = 1$ only
D. For all x
E. For no x
F. For $x = 0$ only

$\{v_1, v_2, v_3\}$ is l.i. iff

$\operatorname{rank} \underbrace{\begin{bmatrix} -1 & 1 & 1 \\ 0 & 1 & x \\ 1 & 2 & 2 \end{bmatrix}}_{A}$ is 3. But

$\operatorname{rank} A = \operatorname{rank} \begin{bmatrix} -1 & 1 & 1 \\ 0 & 1 & x \\ 0 & 3 & 3 \end{bmatrix} = \operatorname{rank} \begin{bmatrix} -1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & x-1 \end{bmatrix} = 3$

iff $x \neq 1$

3. Suppose $A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$. Which one of the following statements is true ?

A. A^{-1} does not exist.
B. The third row of A^{-1} is $[0 \quad 0 \quad 1]$.
C. The second row of A^{-1} is $[-2 \quad 1 \quad 0]$.
D. The first row of A^{-1} is $[1 \quad -3 \quad 2]$.
E. The second column of A^{-1} is $[0 \quad 2 \quad -1]^t$.
F. All of B, C, D, E are true.

$$[A|I_3] = \left[\begin{array}{ccc|ccc} 1 & -2 & 3 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|ccc} 1 & -2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -3 & 2 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{array}\right]$$

4. Let

$$A = \begin{bmatrix} 1 & 1 & -1 & -1 \\ 0 & 0 & 1 & 3 \\ -3 & -3 & 5 & 9 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & -1 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 2 & 6 \end{bmatrix}$$

(a) Find a basis for the row space of A.

(b) Find a basis for the column space of A.

(c) Find a basis for $\ker A = \{x \in \mathbf{R}^4 \mid Ax = 0\}$.

(d) Find the dimension of $\{Ax \mid x \in \mathbf{R}^4\}$.

$$\sim \begin{bmatrix} \textcircled{1} & 1 & 0 & 4 \\ 0 & 0 & \textcircled{1} & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(a) By computations above, $\{(1,1,0,4), (0,0,1,3)\}$ is a basis for row(A)

(b) By computations above (and using the col. space algorithm), $\{(1,0,-3), (-1,1,3)\}$ is a basis for col A

(c) $[A|0] \sim \begin{bmatrix} \textcircled{1} & 1 & 0 & 4 \\ 0 & 0 & \textcircled{1} & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$; $\begin{aligned} x_1 &= -s - 4t \\ x_2 &= s \\ x_3 &= -3t \\ x_4 &= t \end{aligned}$; $s, t \in \mathbb{R}$

So $\ker A = \{ s(-1,1,0,0) + t(-4,0,-3,1) \mid s,t \in \mathbb{R} \}$.

Hence $\{(-1,1,0,0), (-4,0,-3,1)\}$ is a basis of $\ker A$

(d) The subspace $\{Ax \mid x \in \mathbb{R}^4\}$ is col A. But

$\dim \text{col} A = \text{rank} A = 2$. (Or. by (b))

5. Let $u_1 = (0,1,1,0), u_2 = (0,0,0,1), u_3 = (0,1,-1,0)$, and $U = \text{span}\{u_1, u_2, u_3\}$.

a) Show that $\{u_1, u_2, u_3\}$ is an orthogonal set. $\left(U = \{(x,y,z,w) \in \mathbb{R}^4 \mid x = 0\}\right)$

b) Briefly explain why $\{u_1, u_2, u_3\}$ is a basis of U.

c) Find the best approximation to $(1, -1, 2, -1)$ by vectors in U.

d) Extend $\{u_1, u_2, u_3\}$ to a basis of $\mathbf{R}^4$.

a) A short computation shows $u_1 \cdot u_2 = u_1 \cdot u_3 = u_2 \cdot u_3 = 0$; Moreover none of u_1, u_2 or u_3 is zero. Hence $\{u_1, u_2, u_3\}$ is orthogonal

b) Every orthogonal set is l.i., so $\{u_1, u_2, u_3\}$ is l.i., and spans U (by def^n of U). Hence $\{u_1, u_2, u_3\}$ is a basis of U.

c) Let $v = (1, -1, -2, 1)$. We compute $\text{proj}_U v$:

$$\text{proj}_U v = \frac{u_1 \cdot v}{\|u_1\|^2} u_1 + \frac{u_2 \cdot v}{\|u_2\|^2} u_2 + \frac{u_3 \cdot v}{\|u_3\|^2} u_3$$

$$= \frac{1}{2} u_1 + \frac{(-1)}{1} u_2 + \frac{(-3)}{2} u_3$$

$$= (0, -1, 2, -1)$$

d) If $w \in \mathbb{R}^4$, $\{u_1, u_2, u_3, w\}$ is a basis for $\mathbb{R}^4$ iff

$$\text{rank} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ w \end{bmatrix} = 4. \text{ But rank} \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & -1 & 0 \\ & & w & \end{bmatrix} = \text{rank} \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 \\ & & w & \end{bmatrix}$$

$= 4$ if $w = (1, 0, 0, 0)$.

6(a). State whether the following is true or false, and justify your answer.

If A and B are 2×2 matrices such that the columns of AB are linearly independent, then the columns B must also be linearly independent.

TRUE If the columns of AB are l.i., then AB is invertible, so A and B are invertible. Hence the columns of B are also l.i.

6(b). Let A be a $n \times n$ matrix with real entries. Give three additional statements equivalent to

"A is not invertible"

(i) rank $A < n$

(ii) The rows of A are linearly dependent

(iii) $Ax = 0$ has infinitely many solns

(There are just 3 possible answers. See your note for another 9.)